Aarya Vijay Arban
Batch – S11
Roll no. – 07

Aim – Implementation of Stack and Queue using Singly Linked List

Code (Stack):

```
#include <stdio.h>

#include <stdlib.h>

struct node {

   int info;

   struct node *ptr;

}*top,*top1,*temp;


int count = 0;

void push(int data) {

    if (top == NULL)

   {

      top =(struct node *)malloc(1*sizeof(struct node));

      top->ptr = NULL;

      top->info = data;

   }

   else

   {

      temp =(struct node *)malloc(1*sizeof(struct node));

      temp->ptr = top;

      temp->info = data;

      top = temp;

   }

   count++;
```

```
    printf("Node is Inserted\n\n");
}

int pop() {
     top1 = top;

    if (top1 == NULL)
    {
        printf("\nStack Underflow\n");
        return -1;
    }
    else
        top1 = top1->ptr;
    int popped = top->info;
    free(top);
    top = top1;
    count--;
    return popped;
}

void display()
{
    top1 = top;

    if (top1 == NULL)
    {
        printf("\nStack Underflow\n");
        return;
    }
```

```
    printf("The stack is \n");
    while (top1 != NULL)
    {
        printf("%d--->", top1->info);
        top1 = top1->ptr;
    }
    printf("NULL\n\n");

}

int main() {
    int choice, value;
    printf("\nImplementation of Stack using Linked List\n");
    while (1) {
        printf("\n1. Push\n2. Pop\n3. Display\n4. Exit\n");
        printf("\nEnter your choice : ");
        scanf("%d", &choice);
        switch (choice) {
        case 1:
            printf("\nEnter the value to insert: ");
            scanf("%d", &value);
            push(value);
            break;
        case 2:
            printf("Popped element is :%d\n", pop());
            break;
        case 3:
            display();
```

```
            break;
        case 4:
            exit(0);
            break;
        default:
            printf("\n Wrong Choice\n");
        }
    }
}
```

Output:

Implementation of Stack using Linked List

1. Push

2. Pop

3. Display

4. Exit

Enter your choice : 1

Enter the value to insert: 11

Node is Inserted

1. Push

2. Pop

3. Display

4. Exit

Enter your choice : 1

Enter the value to insert: 12

Node is Inserted

1. Push
2. Pop
3. Display
4. Exit

Enter your choice : 3

The stack is

12--->11--->NULL

1. Push
2. Pop
3. Display
4. Exit

Enter your choice : 2

Popped element is :12

1. Push
2. Pop
3. Display
4. Exit

Enter your choice : 3

The stack is

11--->NULL

1. Push

2. Pop

3. Display

4. Exit

Enter your choice : 4

Code (Queue):

```
#include < stdio.h >
#include < stdlib.h >
struct node {
   int data;
   struct node * next;
};

struct node * front = NULL;
struct node * rear = NULL;
void enqueue(int value) {
   struct node * ptr;
   ptr = (struct node * ) malloc(sizeof(struct node));
   ptr - > data = value;
   ptr - > next = NULL;
   if ((front == NULL) && (rear == NULL)) {
      front = rear = ptr;
   } else {
      rear - > next = ptr;
      rear = ptr;
```

```
    }
    printf("Node is Inserted\n\n");
}
int dequeue() {
    if (front == NULL) {
        printf("\nUnderflow\n");
        return -1;
    } else {
        struct node * temp = front;
        int temp_data = front - > data;
        front = front - > next;
        free(temp);
        return temp_data;
    }
}
void display() {
    struct node * temp;
    if ((front == NULL) && (rear == NULL)) {
        printf("\nQueue is Empty\n");
    } else {
        printf("The queue is \n");
        temp = front;
        while (temp) {
            printf("%d--->", temp - > data);
            temp = temp - > next;
        }
        printf("NULL\n\n");
    }
}
```

```
int main() {
    int choice, value;
    printf("\n Implementation of Queue using Linked List\n");
    while (choice != 4) {
        printf("1.Enqueue\n2.Dequeue\n3.Display\n4.Exit\n");
        printf("\nEnter your choice : ");
        scanf("%d", & choice);
        switch (choice) {
            case 1:
                printf("\nEnter the value to insert: ");
                scanf("%d", & value);
                enqueue(value);
                break;
            case 2:
                printf("Popped element is :%d\n", dequeue());
                break;
            case 3:
                display();
                break;
            case 4:
                exit(0);
                break;
            default:
                printf("\n Wrong Choice\n");
        }
    }
    return 0;
}
```

Output:

Implementation of Queue using Linked List

1. Enqueue
2. Dequeue
3. Display
4. Exit

Enter your choice: 1

Enter the value to insert: 12

Node is Inserted

1.Enqueue

2.Dequeue

3.Display

4.Exit

Enter your choice: 1

Enter the value to insert: 45

Node is Inserted

1.Enqueue

2.Dequeue

3.Display

4.Exit

Enter your choice: 1

Enter the value to insert: 56

Node is Inserted

1.Enqueue

2.Dequeue

3.Display

4.Exit

Enter your choice : 3

The queue is

12--->45--->56--->NULL